Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The position of point $(1, 2)$ with respect to Hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$ is -

(A) lies inside hyperbola

(B) lies on hyperbola

(C) lies on line $y = x$

(D) lies outside hyperbola

Q.2

A hyperbola having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$ Then its equation is

(A) $x^2\,\text{cosec}^2\,\theta - y^2\sec^2\theta = 1$

(B) $x^2\sec^2\theta - y^2\,\text{cosec}^2\,\theta = 1$

(C) $x^2\sin^2\theta - y^2\cos^2\theta = 1$

(D) $x^2\cos^2\theta - y^2\sin^2\theta = 1$

Q.3

If AB is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that $\Delta\mathrm{OAB}$ is an equilateral triangle O being the origin, then the eccentricity of the hyperbola satisfies

(A) $\mathrm{e} > \sqrt{3}$

(B) $1 < \mathrm{e} < \frac{2}{\sqrt{3}}$

(C) $\mathrm{e} = \frac{2}{\sqrt{3}}$

(D) $\mathrm{e} > \frac{2}{\sqrt{3}}$

Q.4

Equation of the latus rectum of the hyperbola $(10x - 5)^2 + (10y - 2)^2 = 9(3x + 4y - 7)^2$ is

(A) $y - \frac{1}{5} = -\frac{3}{4}\left(x - \frac{1}{2}\right)$

(B) $x - \frac{1}{5} = -\frac{3}{4}\left(y - \frac{1}{2}\right)$

(C) $y + \frac{1}{5} = -\frac{3}{4}\left(x + \frac{1}{2}\right)$

(D) $x + \frac{1}{5} = -\frac{3}{4}\left(y + \frac{1}{2}\right)$

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Q.5

The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is :

(A) $\frac{4}{3}$

(B) $\frac{4}{\sqrt{3}}$

(C) $\frac{2}{\sqrt{3}}$

(D) None of these

Q.6

If the eccentricity of the hyperbola $x^2 - y^2 \sec^2\theta = 4$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2 \sec^2\theta + y^2 = 16$, then the value of θ equals

(A) $\frac{\pi}{6}$

(B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Q.7

Equation of the hyperbola whose directrix is $2x + y = 1$, focus $(1, 2)$ and eccentricity $\sqrt{3}$ is

(A) $7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$

(B) $5x^2 - 2y^2 + 10xy + 2x + 5y - 20 = 0$

(C) $4x^2 + 8y^2 + 8xy + 2x - 2y + 10 = 0$

(D) None of these

Q.8

What are the points of intersection of the curve $4x^2 - 9y^2 = 1$ with its conjugate axis?

(A) $(1/2, 0)$ and $(-1/2, 0)$

(B) $(0, 2)$ and $(0, -2)$

(C) $(0, 3)$ and $(0, -3)$

(D) No such point exists

Q.9

The curve described parametrically by $x = 2 - 3\sec t, y = 1 + 4\tan t$ represents :

(A) An ellipse centred at $(2, 1)$ and of eccentricity $\frac{3}{5}$

(B) A circle centred at $(2, 1)$ and of radius 5 units

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(C) A hyperbola centred at $(2,1)$& of eccentricity $\frac{8}{5}$

(D) A hyperbola centred at $(2,1)$& of eccentricity $\frac{5}{3}$

Q.10

A hyperbola having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2+4y^2=12$. Then its equation is

(A) $x^2\operatorname{cosec}^2\theta-y^2\sec^2\theta=1$

(B) $x^2\sec^2\theta-y^2\operatorname{cosec}^2\theta=1$

(C) $x^2\sin^2\theta-y^2\cos^2\theta=1$

(D) $x^2\cos^2\theta-y^2\sin^2\theta=1$

Q.11

The equation $\sqrt{(x-2)^2+(y-1)^2}-\sqrt{(x+2)^2+y^2}=c$ will represent a hyperbola if

(A) $c\in(0,\ 6)$

(B) $c\in(6,\ 5)$

(C) $c\in\left(0,\ \sqrt{17}\right)$

(D) none of these

Q.12

The foci of the hyperbola $9x^2-16y^2=144$ are

(A) $(\pm4,0)$

(B) $(0,\pm4)$

(C) $(\pm5,0)$

(D) $(0,\pm5)$

Q.13

The distance between the foci of a hyperbola is double the distance between its vertices and the length of its conjugate axis is 6 . The equation of the hyperbola referred to its axes as axes of co-ordinates is

(A) $3x^2-y^2=3$

(B) $x^2-3y^2=3$

(C) $3x^2-y^2=9$

(D) $x^2-3y^2=9$

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Q.14

The coordinates of the foci of the rectangular hyperbola $xy = c^2$ are

(A) $(\pm c, \pm(c)$

(B) $(\pm c\sqrt{2}, \pm c\sqrt{2})$

(C) $\left(\pm\frac{c}{\sqrt{2}}, \pm\frac{c}{\sqrt{2}}\right)$

(D) None of these

Q.15

The latus rectum of the hyperbola $9x^2 - 16y^2 - 18x - 32y - 151 = 0$

(A) $\frac{9}{4}$

(B) 9

(C) $\frac{3}{2}$

(D) $\frac{9}{2}$

Q.16

The auxiliary equation of circle of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, is

(A) $x^2 + y^2 = a^2$

(B) $x^2 + y^2 = b^2$

(C) $x^2 + y^2 = a^2 + b^2$

(D) $x^2 + y^2 = a^2 - b^2$

Q.17

Let $z = 1 - t + i\sqrt{t^2 + t + 2}$, where t is a real parameter. The locus of z in the argand plane is

(A) an ellipse

(B) hyperbola

(C) a straight line

(D) None of thees

Q.18

The one which does not represent a hyperbola is

(A) $xy = 1$

(B) $x^2 - y^2 = 5$

(C) $(x - 1)(y - 3) = 3$

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(D) $x^2 - y^2 = 0$

Q.19

The eccentricity of the hyperbola can never be equal to

(A) $\sqrt{\frac{9}{5}}$

(B) $2\sqrt{\frac{1}{9}}$

(C) $3\sqrt{\frac{1}{8}}$

(D) 2

Q.20

If $x + y = k$ is a tangent to the hyperbola x^2 - $2y^2 = 18$, then sum of squares of possible values of k is

(A) 3

(B) 9

(C) 12

(D) 18

Answers & Solutions

Q.1 Answer:

lies outside hyperbola

Solution:

For $(1, 2)$ **H=**$\frac{x^2}{4} - \frac{y^2}{9} - 1 < 0$

$\Rightarrow$ **since** $H < 0 \Rightarrow$ **this implies pt lies outside hyperbola.**

Q.2 Answer:

$x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

Solution:

Equation of the ellipse is $3x^2 + 4y^2 = 12 \Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$ **...(1)**

Eccentricity $e_1 = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$

So, the foci of ellipse are $(1, 0)$ **and** $(-1, 0)$ **Let the equation of the required hyperbola be** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **. . . . . (2)**

Given $2a = 2 \sin \theta \Rightarrow a = \sin \theta$

Since the ellipse (1) **and the hyperbola** (2) **are confocal, so the foci of hyperbola are** $(1, 0)$ **and** $(-1, 0)$ **too. If the eccentricity, of hyperbola be** e_2 **then** $ae_2 = 1 \Rightarrow \sin \theta e_2 = 1 \Rightarrow e_2 = \operatorname{cosec} \theta$

$b^2 = a^2 \left(e^2 - 1\right) = \sin^2 \theta \left(\operatorname{cosec}^2 \theta - 1\right) = \cos^2 \theta$

Required equation of the hyperbola is $\frac{x^2}{\sin^2 \theta} - \frac{y^2}{\cos^2 \theta} = 1 \Rightarrow x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

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Q.3 Answer:

$e > \frac{2}{\sqrt{3}}$

Solution:

Let the length of the double ordinate be 21

$\therefore AB = 2l$ **and** $AM = BM = 1$

Clearly ordinate of point A **is** $1.$

The abscissa of the point A **is given by**

$\frac{x^2}{a^2} - \frac{1^2}{b^2} = 1 \Rightarrow x = \frac{a\sqrt{b^2+1^2}}{b}$

$\therefore A$ is $\left(\frac{a\sqrt{b^2+1^2}}{b}, l\right)$

Since $\triangle OAB$ **is equilateral triangle, therefore**

$OA = AB = OB = 2l$

Also, $OM^2 + AM^2 = OA^2 \Rightarrow \frac{a^2\left(b^2+l^2\right)}{b^2} + l^2 = 4l^2$

$\therefore 1^2 = \frac{a^2 b^2}{3b^2 - a^2}$

Since $l^2 > 0$

$\therefore \frac{a^2 b^2}{3b^2-a^2} > 0 \Rightarrow 3b^2 - a^2 > 0 \Rightarrow 3a^2\left(e^2 - 1\right) - a^2 > 0 \Rightarrow e > \frac{2}{\sqrt{3}}$

Q.4 Answer:

$y - \frac{1}{5} = -\frac{3}{4}\left(x - \frac{1}{2}\right)$

Solution:

$\Rightarrow \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{5}\right)^2 = \frac{9}{4}\left(\frac{3x+4y-7}{5}\right)^2$

$\Rightarrow \sqrt{\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{5}\right)^2} = \frac{3}{2}\left(\frac{3x+4y-7}{5}\right) \ldots\ldots\ldots(2)$

LHS is the distance from the point $\left(\frac{1}{2}, \frac{1}{5}\right)$ **and RHS is** $\frac{3}{2}$ **times distance from** $3x + 4y - 7 = 0$

so equation 2 is focus-directrix definition of Hyperbola with $e = \frac{3}{2}$**, focus at** $\left(\frac{1}{2}, \frac{1}{5}\right)$ **and** $3x + 4y - 7 = 0$ **is a directrix.**

The latus rectum through this focus is parallel to Directrix. So it's equation is then

$3\left(x - \frac{1}{2}\right) + 4\left(y - \frac{1}{5}\right) = 0$

Q.5 Answer:

$\frac{2}{\sqrt{3}}$

Solution:

The standard equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Latus rectum $= \frac{2b^2}{a}$**, Conjugate axis** $= 2b$**, Distance between the foci** $= 2$ **ae According to the question,**

$\frac{2b^2}{a} = 8$ **...(i)**

$2b = \frac{1}{2}(2ae) \Rightarrow b = \frac{ae}{2}$ **...(ii)**

$\frac{2}{a}\left(\frac{ae}{2}\right)^2 = 8$ **[from (i) and (ii)]**

$\Rightarrow 2 \cdot \frac{a^2e^2}{4a} = 8$

$\Rightarrow \quad ae^2 = 16$

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From (i), $b^2 = 4a$ **Using** $b^2 = a^2\left(e^2-1\right)$ **we get** $(4a) = a^2\left(e^2-1\right) \Rightarrow 4 = \frac{16}{e^2}\left(e^2-1\right)$
$\Rightarrow 16 - \frac{16}{e^2} = 4 \Rightarrow \frac{16}{e^2} = 12 \quad \therefore \quad e = \frac{2}{\sqrt{3}}$

Q.6 Answer:

$\frac{3\pi}{4}$

Solution:

Given: $x^2 - y^2\sec^2\theta = 4$ **and**
$x^2\sec^2\theta + y^2 = 16$
$\Rightarrow \frac{x^2}{4} - \frac{y^2}{4\cos^2\theta} = 1$ **and** $\frac{x^2}{16\cos^2\theta} + \frac{y^2}{16} = 1$
According to problem $\frac{4+4\cos^2\theta}{4} = 3\left(\frac{16-16\cos^2\theta}{16}\right)$
$\Rightarrow 1 + \cos^2\theta = 3\left(1-\cos^2\theta\right) \Rightarrow 4\cos^2\theta = 2$
$\Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$

Q.7 Answer:

$7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$

Solution:

Let $P(x,y)$ **be any point on the hyperbola and** PM **is perpendicular from** P **on the directrix, Then by definition,** $SP = ePM \Rightarrow (SP)^2 = e^2(PM)^2$
$\Rightarrow (x-1)^2 + (y-2)^2 = 3$
$\left\{\frac{2x+y-1}{\sqrt{4+1}}\right\}^2 (\because e = \sqrt{3})$
$\Rightarrow 5\left(x^2+y^2-2x-4y+5\right)$
$= 3\left(4x^2+y^2+1+4xy-2y-4x\right)$
$\Rightarrow 7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$
Which is the required hyperbola.

Q.8 Answer:

No such point exists

Solution:

The given equation of curve is $4x^2 - 9y^2 = 1$
$\Rightarrow \quad \frac{x^2}{1/4} - \frac{y^2}{1/9} = 1$
This is an equation of a hyperbola which does not intersect with conjugate axes. Hence, no point of intersection exists.

Q.9 Answer:

A hyperbola centred at $(2,1)\&$ of
eccentricity $\frac{5}{3}$

Solution:

Given, $x = 2 - 3\sec t, y = 1 + 4\tan t$
$\Rightarrow \sec t = \frac{x-2}{-3}, \tan t = \frac{y-1}{4}$

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Since, $\sec^2 t - \tan^2 t = 1$

$\frac{(x-2)^2}{9} - \frac{(y-1)^2}{16} = 1$

which is a hyperbola with centre at $(2, 1)$ **and**

eccentricity e, given by $16 = 9\left(e^2 - 1\right)$

$\Rightarrow 9e^2 = 25 \Rightarrow e^2 = \frac{25}{9} \Rightarrow e = \frac{5}{3}$

Q.10 Answer:

$x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$

Solution:

Equation of the ellipse is $3x^2 + 4y^2 = 12 \Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$

Eccentricity $e_1 = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$

So, the foci of ellipse are $(1, 0)$ **and** $(-1, 0)$ **Let the equation of the required hyperbola be** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Given $2a = 2\sin\theta \Rightarrow a = \sin\theta$

Since the ellipse (1) and the hyperbola (2) are confocal, so

the foci of hyperbola are $(1, 0)$ **and** $(-1, 0)$ **too. If the eccentricity, of hyperbola be** e_2 **then** $ae_2 = 1 \Rightarrow \sin\theta e_2 = 1 \Rightarrow e_2 = \operatorname{cosec}\theta$

$\therefore b^2 = a^2\left(e_2^2 - 1\right)^2 = \sin^2\theta^2\left(\operatorname{cosec}^2\theta - 1\right) = \cos^2\theta$

$\therefore$ **Required equation of the hyperbola is**

$\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1 \Rightarrow x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$

Q.11 Answer:

$c \in \left(0, \sqrt{17}\right)$

Solution:

Coordinates of focus are $S_1\left(2, 1\right)$ **and** $S_2\left(-2, 0\right)$ **and** $P\left(x, y\right)$ **for hyperbola** $PS_1 - PS_2 = c$

$\Rightarrow c < S_1 S_2 \Rightarrow 0 < c < \sqrt{17}$

Q.12 Answer:

$(\pm 5, 0)$

Solution:

The equation of hyperbola is $\frac{x^2}{16} - \frac{y^2}{9} = 1$ Now $b^2 = a^2\left(e^2 - 1\right) \Rightarrow e = \frac{5}{4}$ Hence foci are $(\pm ae, 0) \triangleright \left(\pm 4 \cdot \frac{5}{4}, 0\right)$ i.e., $(\pm 5, 0)$

Q.13 Answer:

$3x^2 - y^2 = 9$

Solution:

$\frac{2ac}{2} = 2a \Rightarrow e = 2$

$e^2 = 4 = 1 + \frac{b^2}{a^2} \quad \ldots(1)$

Also, $2b = 6$

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$\Rightarrow b = 3$

From (I) & (II)

$1 + \frac{(3)^2}{a^2} = 4$

$a^2 = 3$

$\Rightarrow \frac{x^2}{3} - \frac{y^2}{9} = 1 \quad \Rightarrow \quad 3x^2 - y^2 = 9$

Q.14 Answer:

$(\pm c\sqrt{2}, \pm c\sqrt{2})$

Solution:

$xy = c^2$ **as** $c^2 = \frac{a^2}{2}$**. Here, co-ordinates of focus are** $(ae\cos 45^\circ, ae\sin 45^\circ) \equiv (c\sqrt{2}, c\sqrt{2}), \{\because e = \sqrt{2}, a = c\sqrt{2}\}$ **Similarly other focus is** $(-c\sqrt{2}, -c\sqrt{2})$ **Note: Students should remember this question as a fact.**

Q.15 Answer:

$\frac{9}{2}$

Solution:

$$9x^2 - 18x + 9 - 16y^2 - 32y - 16 = 144 \Rightarrow \frac{(x-1)^2}{16} - \frac{(y+1)^2}{9} = 1$$

$$\text{P Latus rectum } = \frac{2b^2}{a} = \frac{2 \times 9}{4} = \frac{9}{2}$$

Q.16 Answer:

$x^2 + y^2 = a^2$

Solution:

The equation is $(x-0)^2 + (y-0)^2 = a^2$**.**

Q.17 Answer:

hyperbola

Solution:

Let $z = 1 - t + i\sqrt{t^2 + t + 2}$

We know that $z = x +$ **iy** $x + iy = 1 - t + i\sqrt{t^2 + t + 2}$

compare real and imaginary part, we get $x = 1 - t \Rightarrow t = 1 - x$

and $y = \sqrt{t^2 + t + 2}$

$\Rightarrow y^2 = t^2 + t + 2$

$\Rightarrow y^2 = (1-x)^2 + (1-x) + 2$

$\Rightarrow y^2 = 1 + x^2 - 2x + 1 - x + 2$

$\Rightarrow y^2 = x^2 - 3x + 4$

$\Rightarrow y^2 = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}$

Which is a hyperbola.

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Q.18 Answer:

$x^2 - y^2 = 0$

Solution:

For hyperbola $\Delta \neq 0$ **and** $h^2 > ab.$ **Here** $\Delta = 0$

Q.19 Answer:

$2\sqrt{\frac{1}{9}}$

Solution:

Since $e > 1$ **always for hyperbola and** $\frac{2}{3} < 1.$

Q.20 Answer:

18

Solution:

$\frac{x^2}{18} - \frac{y^2}{9} = 1$

Equation of Tangent : $y = -x + k$

$\therefore k^2 = 18(-1)^2 - 9$

$k = \pm 3$

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